Cricket Player Performance Tracking Application: CRICKSTATS

User Story:

As a Salesforce Admin, I want to create a comprehensive player performance tracking application within Salesforce to capture and analyse detailed match-by-match information of cricket players. This application will include custom objects such as Player and Match Performance with carefully defined fields and relationships to ensure data integrity and accessibility. By leveraging master-detail and lookup relationships, the application will enable robust tracking of individual and team performances, facilitate insightful reporting and analytics, and support the overall management of player statistics, thereby enhancing strategic decision-making and player development initiatives.

1. INTRODUCTION

1.1 Project Overview:

CRICKSTATS is a Salesforce-based application designed to track, manage, and analyse cricket players' performance on a match-by-match basis. The system uses custom objects, relationships, dashboards, and automation tools to provide structured performance insights. It enables coaches, analysts, and administrators to record player statistics, generate reports, and support strategic decision-making.

1.2 Purpose:

The main purpose of this project is:

To digitize player performance records

To improve accuracy and accessibility of cricket statistics

To provide real-time analytics

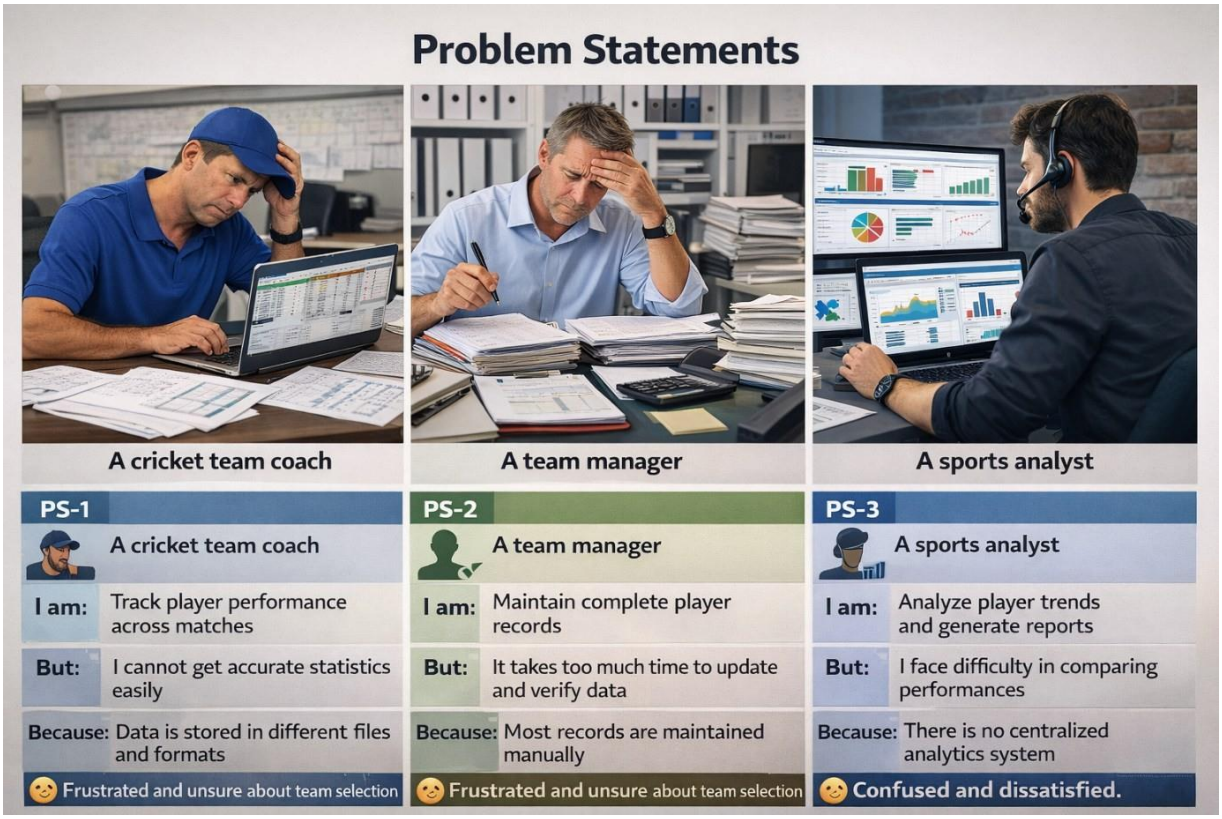
To support player development and team planning

To automate data management using Salesforce

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A cricket team coach	Track player performance across matches	I cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated and unsure about team selection
PS-2	A team manager	Maintain complete player records	It takes too much time to update and verify data	Most records are maintained manually	Stressed and inefficient
PS-3	A sports analyst	Analyze player trends and generate reports	I face difficulty in comparing performances	There is no centralized analytics system	Confused and dissatisfied



2.2 Empathy Map Canvas

Target Users

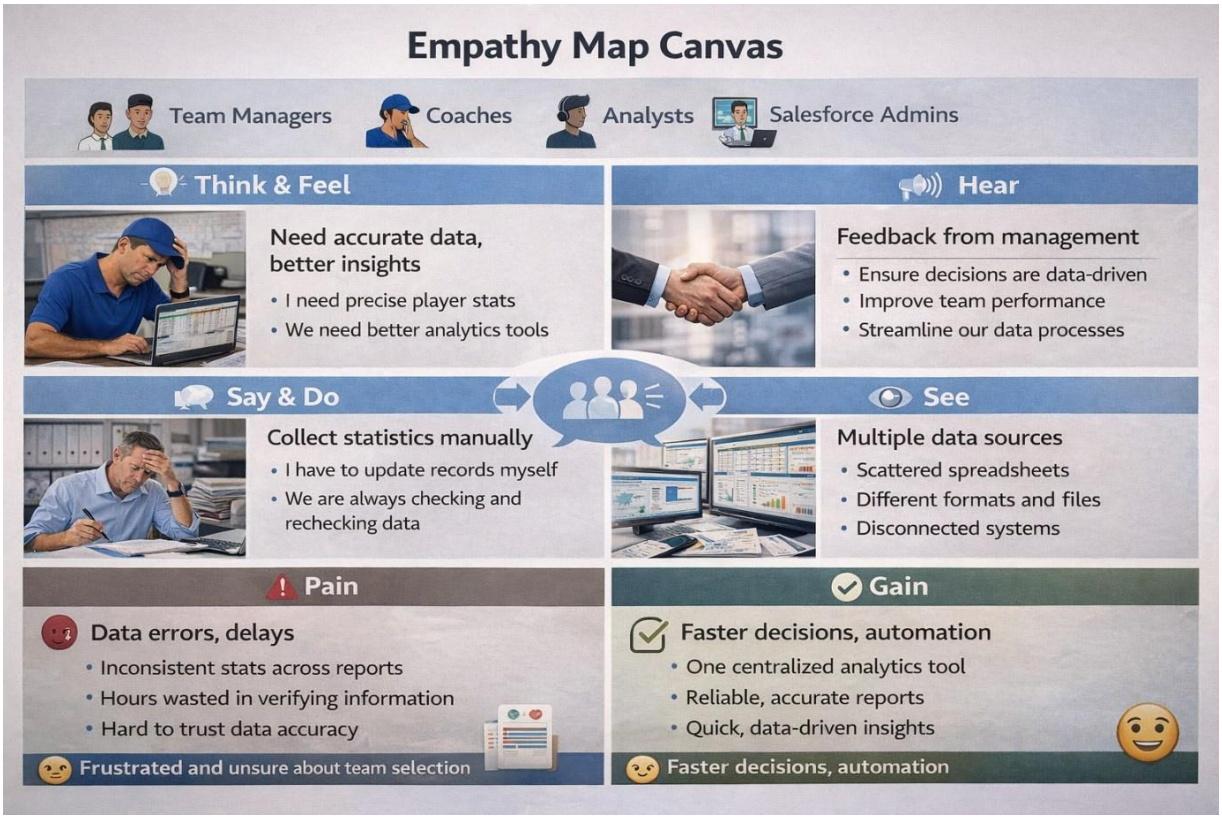
- Team Managers
- Coaches
- Analysts
- Salesforce Admins

Empathy Analysis

Aspect	Description

- **Think & Feel**-- Need accurate data, better insights
- **Hear** -- Feedback from management
- **See** -- Multiple data sources
- **Say & Do** --Collect statistics manually

- **Pain** -- Data errors, delays
- **Gain** -- Faster decisions, automation

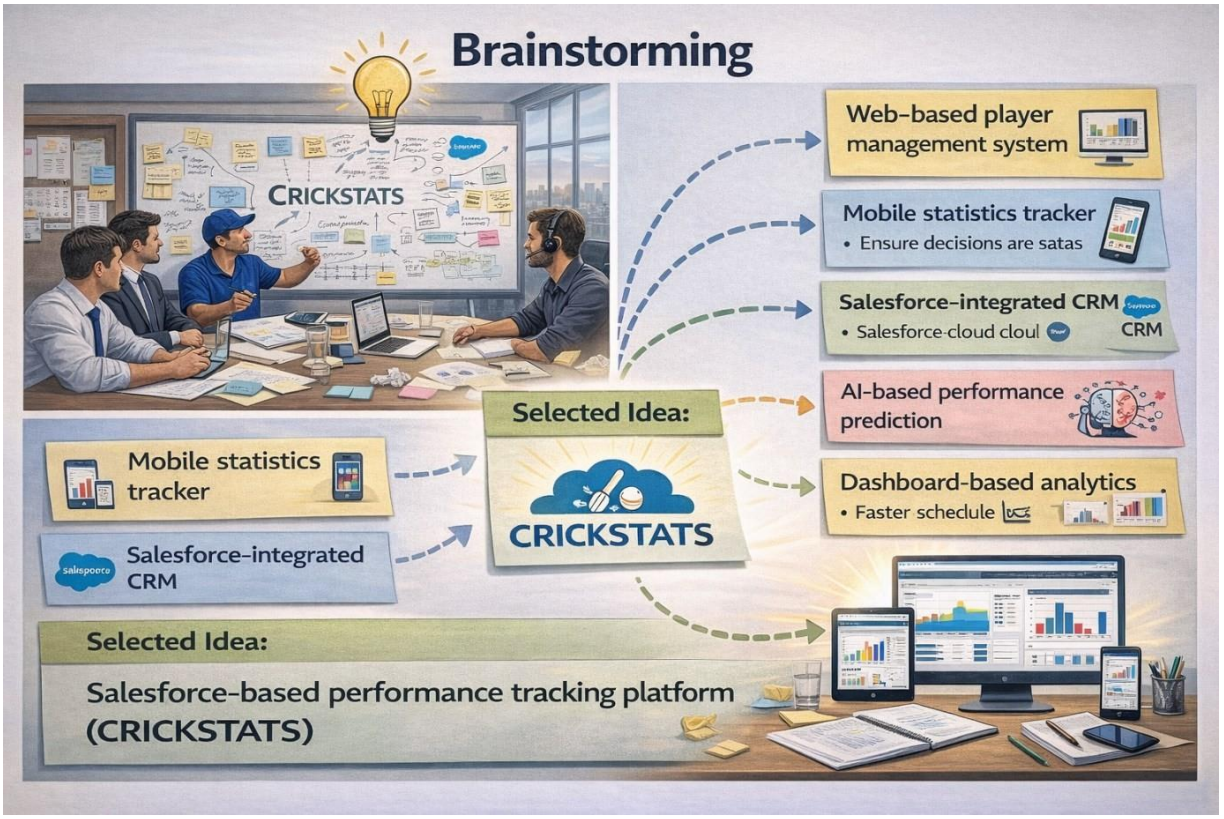


2.3 Brainstorming

Brainstormed Ideas

- Web-based player management system
- Mobile statistics tracker
- Salesforce-integrated CRM
- AI-based performance prediction
- Dashboard-based analytics

Selected Idea: Salesforce-based performance tracking platform (CRICKSTATS)



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

- Login
- Enter Player Data
- Record Match Performance
- Generate Reports
- Analyse Results
- Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Player Registration
FR-2	Match Performance Entry
FR-3	Dashboard Generation
FR-4	Report Creation
FR-5	Flow Automation

Non-Functional Requirements (NFRs) – CRICKSTATS

NFR-1: Usability

The system should be easy to understand and use by coaches, managers, analysts, and

admins without requiring technical training.

Requirements

- Simple and intuitive user interface
- Clear navigation menus
- Easy-to-read dashboards
- Minimal learning curve

Examples:

- A new coach should be able to:
- Log in
- Enter match data
- View player reports
- within a few minutes without special training.

Example: The Salesforce Lightning interface provides drag-and-drop dashboards and user-friendly forms.

NFR-2: Security

The system must protect sensitive player data from unauthorized access.

Requirements

- Role-based access control
- Secure authentication
- Encrypted data storage

- Regular backups

Examples:

- Coaches can view player statistics
- Managers can edit records
- Analysts can generate reports
- Only admins can modify system settings

Example: Salesforce Profiles and Permission Sets restrict access to confidential information.

NFR-3: Reliability

The system should function correctly without failure and store data accurately.

Requirements

- No data loss
- Automatic backups
- Error handling
- System stability

Example:

If a manager enters match statistics and loses internet connection, the data should be saved automatically when reconnected.

Salesforce cloud backup ensures recovery in case of system failure.

NFR-4: Performance

The system should respond quickly and handle large volumes of data efficiently.

Requirements

- Fast page loading

- Quick report generation
- Smooth dashboard rendering
- Minimal response time

Example:

When an analyst generates a performance report for 100+ players, the report should load within 2–3 seconds.

Optimized Salesforce queries and caching improve response speed.

NFR-5: Availability

The system should be accessible whenever users need it.

Requirements

- 24/7 system availability
- Minimal downtime
- Cloud-based access
- Disaster recovery

Example:

Coaches should be able to access CRICKSTATS even during tournaments at night or on weekends.

Salesforce provides high uptime with cloud hosting.

NFR-6: Scalability

The system should support increasing numbers of users, players, and records.

Requirements

- Handle growing datasets
- Support more teams and leagues

- Expand storage capacity
- Maintain performance under load

Example:

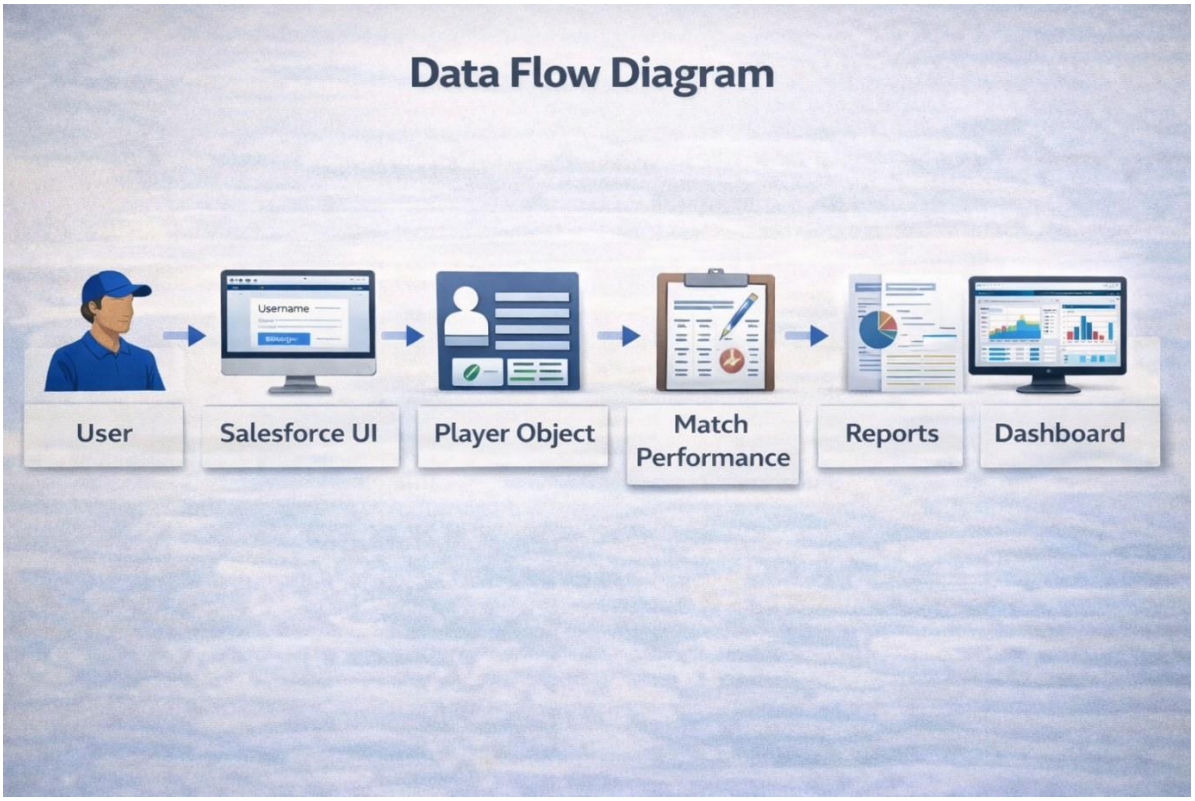
If CRICKSTATS is extended from one team to 50 teams, the system should still perform smoothly.

Salesforce cloud infrastructure automatically scales storage and processing.

3.3 Data Flow Diagram

Data Flow

User → Salesforce UI → Player Object → Match Performance → Reports → Dashboard



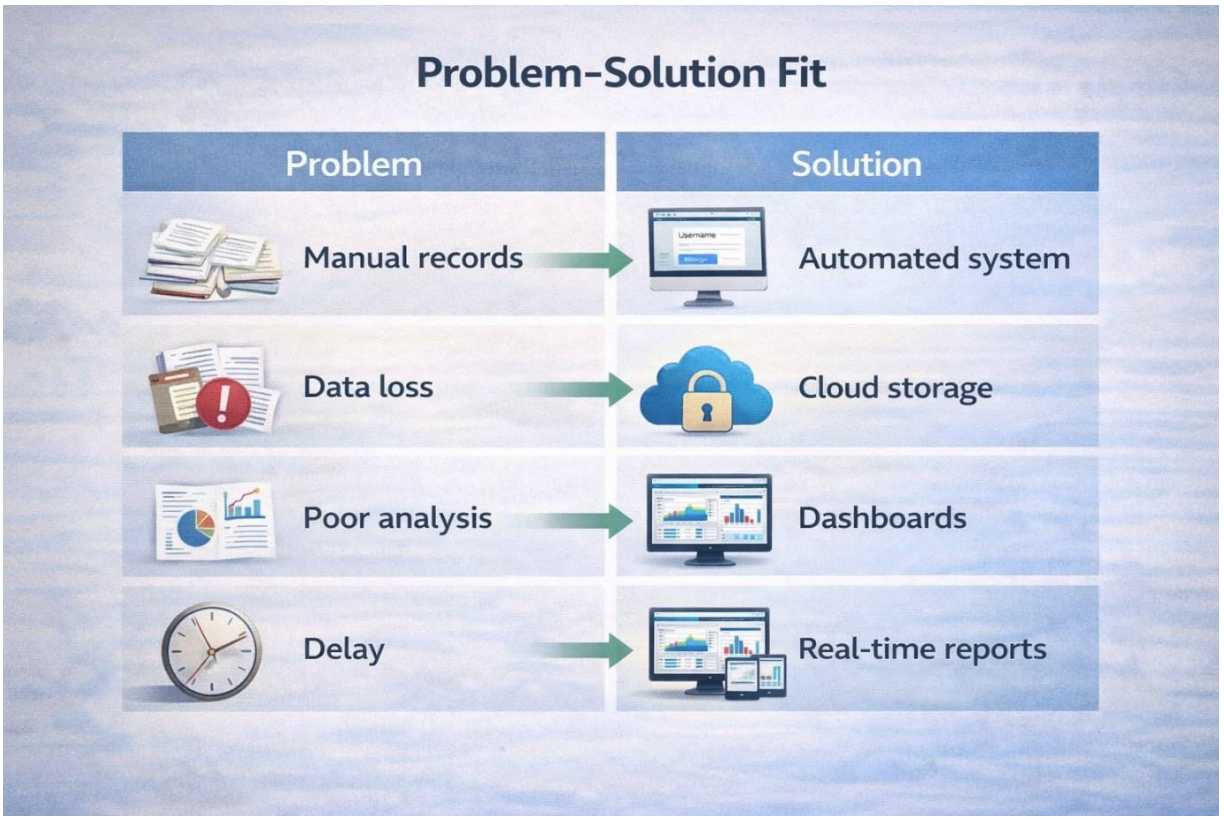
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning
Analytics	Reports & Dashboards
Automation	Salesforce Flows
Backend	Salesforce Objects
Database	Salesforce Cloud

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The CRICKSTATS system effectively addresses the key challenges faced in traditional cricket performance management. Earlier, most player and match data were maintained using manual records, which were time-consuming and prone to errors. This issue has been resolved by implementing an automated system through Salesforce, which ensures accurate and efficient data entry and management. The problem of data loss, caused by improper storage and lack of backups, is overcome by using secure cloud storage that safely stores all information and allows easy retrieval. Poor analysis of player performance, due to scattered and unorganized data, is addressed through interactive dashboards that present clear visual insights and reports. Additionally, delays in accessing performance information and generating reports are eliminated by providing real-time reporting features, enabling users to make faster and more informed decisions. Thus, CRICKSTATS successfully bridges the gap between existing problems and modern technological solutions.



4.2 Proposed Solution

Description

CRICKSTATS provides:

- Custom objects (Player, Match Performance)
- Master-detail relationships
- Roll-up summaries
- Lightning App
- Automated flows
- Interactive dashboards

Key Features

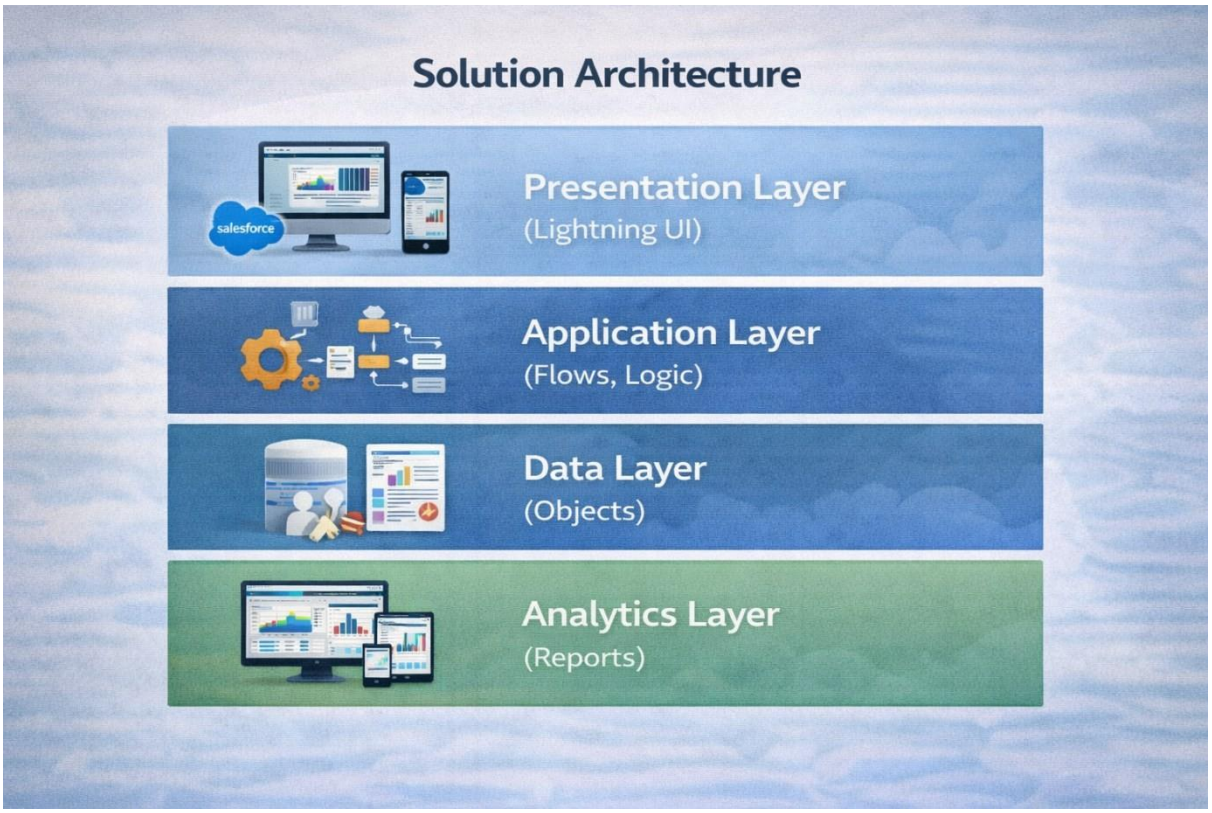
- Centralized database
- Live statistics
- Secure access

- Scalable design
- Easy reporting

4.3 Solution Architecture

Architecture Layers

- Presentation Layer (Lightning UI)
- Application Layer (Flows, Logic)
- Data Layer (Objects)
- Analytics Layer (Reports)



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile-based execution
- Object creation → automation → reporting
- Sprint-wise development

- **Product Backlog, Sprint Schedule, and Estimation (4 Marks)**

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the CRICKSTATS system.	1	High	Member1
Sprint-1	Data Modelling	USN-2	As an admin, I want to create Player and Match Performance objects so that player and match data can be stored systematically.	3	High	Member2
Sprint-2	Navigation	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage records within the application.	3	High	Member3
Sprint-2	UI Design	USN-4	As a coach, I want a dedicated Lightning App so that all performance tracking features are accessible from one dashboard.	3	High	Member4
Sprint-2	Data Structure	USN-5	As an admin, I want to create player and match fields so that complete performance details can be captured accurately.	5	Medium	Member1

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Data Structure	USN-6	As a user, I want customized page layouts so that only relevant fields are displayed during data entry.	5	High	Member2
Sprint-2	Automation	USN-7	As a system, I want to validate player and match inputs so that incorrect data entries are prevented.	3	High	Member4
Sprint-3	Automation	USN-8	As a system, I want automated flows to update player statistics so that performance data remains accurate.	5	High	Member1

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Flow

Milestone 1: Creation of developer account

Milestone 2: Object Creation

Milestone 3: Tabs

Milestone 4: The Lightning App

Milestone 5: Fields & Relationships

Milestone 6: Roll-up Summary

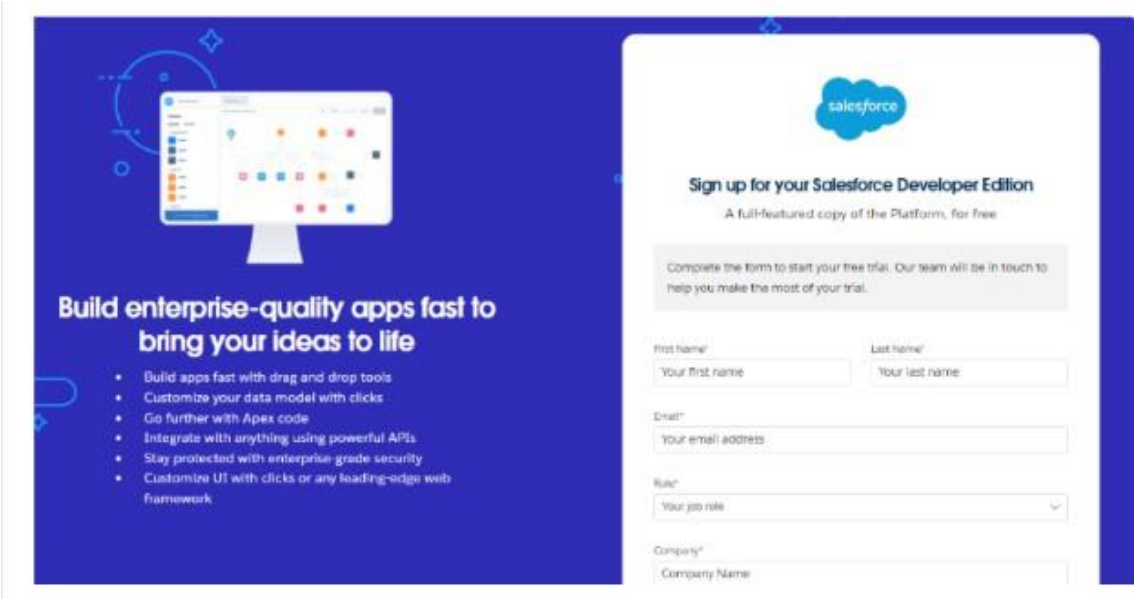
Milestone 7: Flows

Implementation

Milestone 1 - Salesforce developer account creation:

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign-up form, enter the following details:



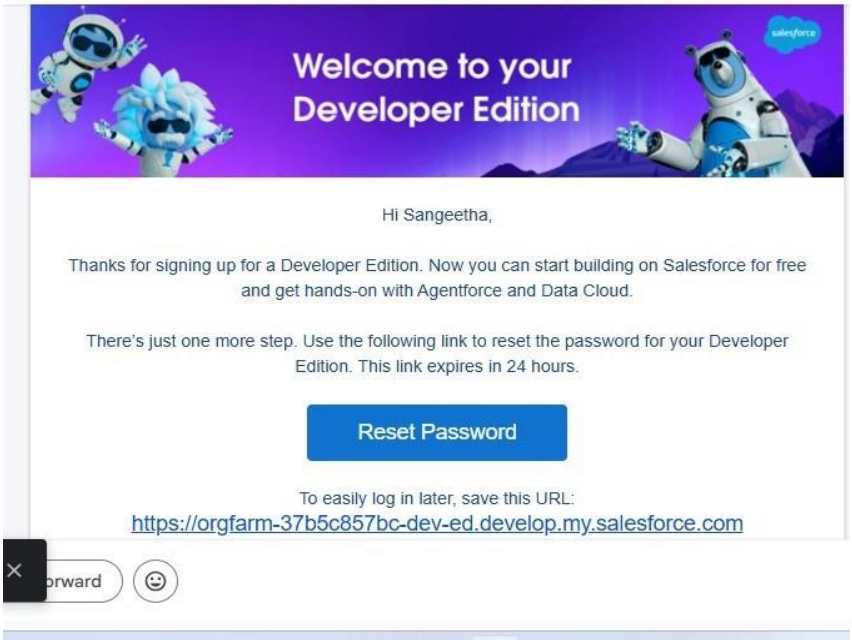
1. First name & Last name
2. Email
3. Role: Developer
4. Company: College or Company Name
5. County: India
6. Postal Code: pin code
7. Username: should be a combination of your name and company

   This need not be an actual email id; you can give anything in that format: username@organization.com

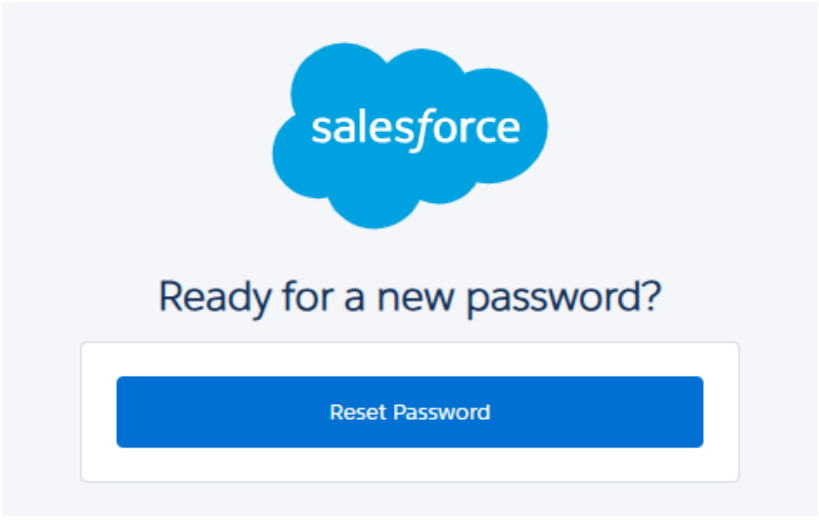
   Click on sign me up after filling these.

Activity 2: Account Activation:

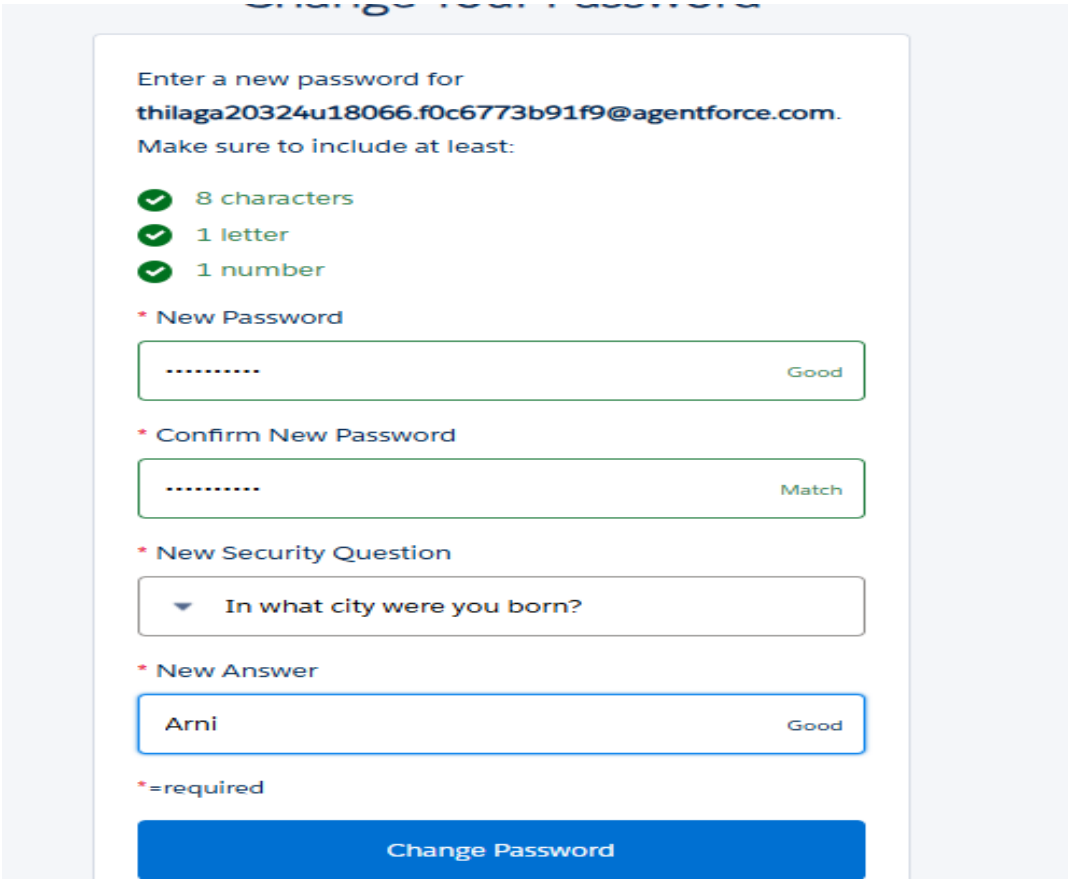
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.
2. Click on Reset password.



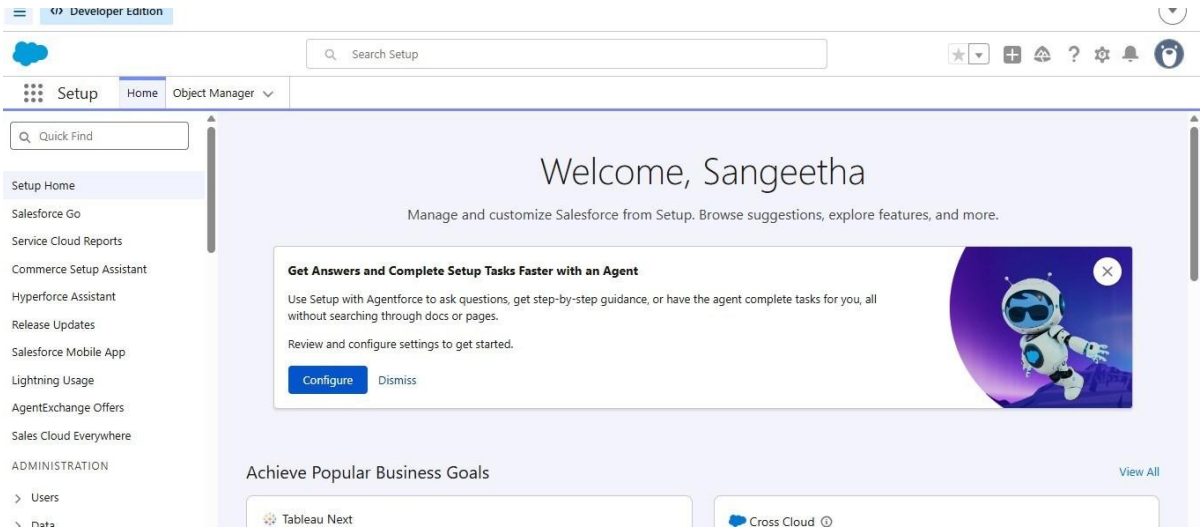
3. Again, click reset password.



4. Give password and answer a security question and click on change password.



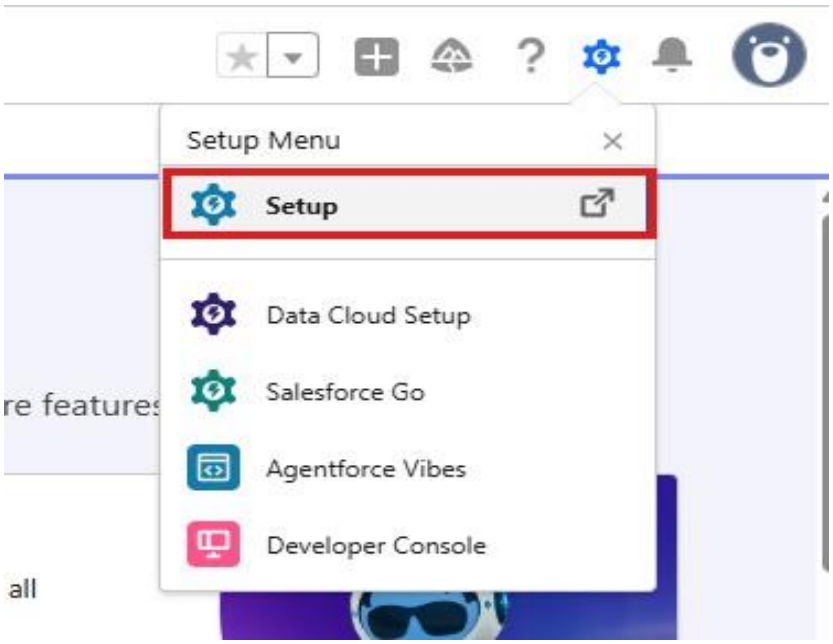
5. Then you will redirect to your salesforce setup page.



Milestone 2 - Object Creation

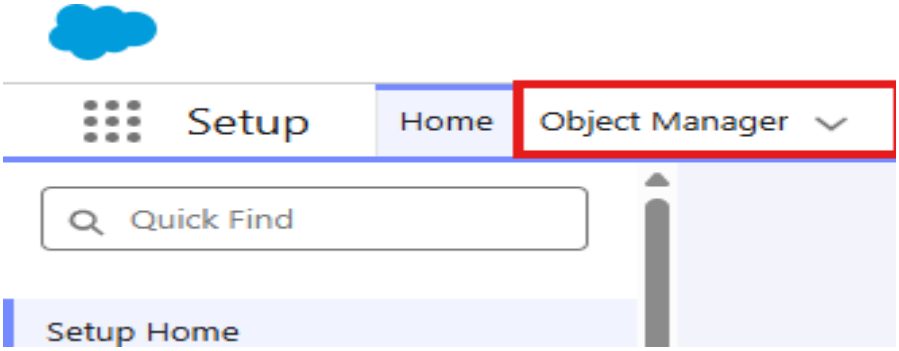
Activity 1: Player Object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.

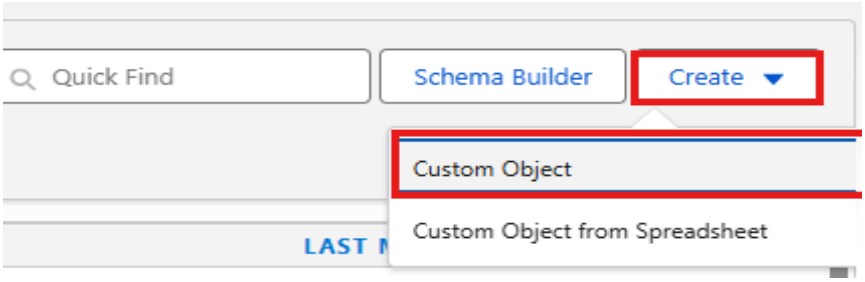


2.

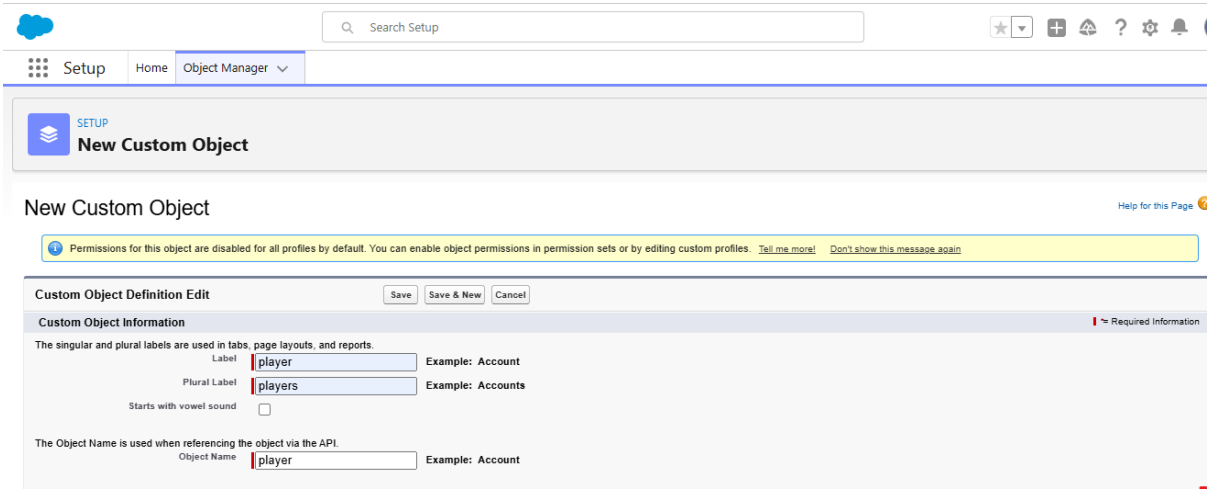
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.



3. On the Object Manager page, click Create | Custom Object.



1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.



- **Label**: Player
- **Plural Label**: Players
- Enter Record Name Label and Format

  Record Name: Player Name

  Data Type**:** Text
- **In Optional features:** check the boxes for Allow Reports | Allow Activities | Track Field History.
- **Search Status:** check the box for Allow Search.
- Save

Activity 2: Match Performance Object

1.From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

- **Label**: Match Performance
- **Plural Label**: Match Performances
- Enter Record Name Label and Format

  Record Name: Customer Account Name

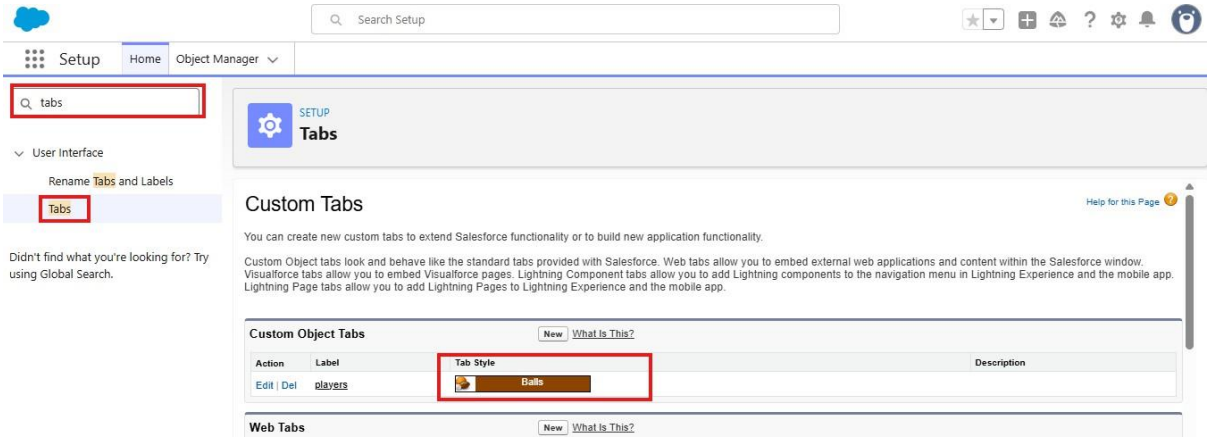
Data Type: Text

- **In Optional features:** check the boxes for Allow Reports | Allow Activities | Track Field History.
- **Search Status:** check the box for Allow Search.
- Save.

Milestone 3 - Tabs

Activity 1: Custom tabs creation

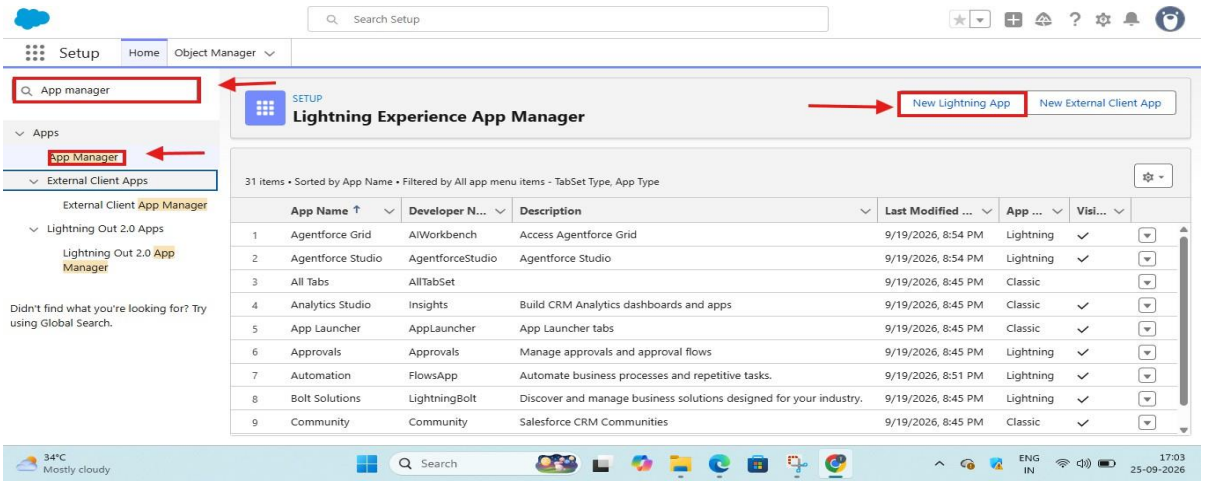
1. Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)
2. Select Object (Player) → Select the tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) uncheck the include tab.
3. Make sure that the Append tab to users' existing personal customizations is checked.
4. Click save.
5. Repeat the same steps for Match Performances.



Milestone 4 - The Lightning App

Activity 1: Create a lightning app

1. Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.



2. Fill the app name in app details and branding as follow

App Name: CRICKSTATS

Developer Name: Auto Populates

Add image optional (if you want to give any image you can otherwise not mandatory) - Add Primary colour Hex or leave it to default.

Then click Next.

3. In App options

Navigation style: Standard navigation

Setup experience: setup

Supporters form factors: Desktop and phone

Then click Next

4. In Utility items

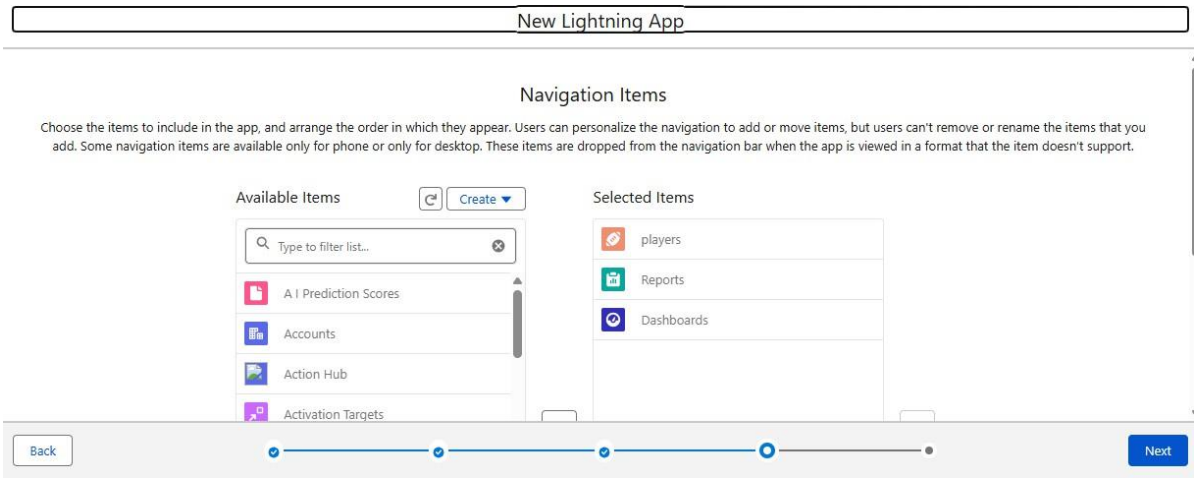
Utility Bar alignment: Default

Then click Next.

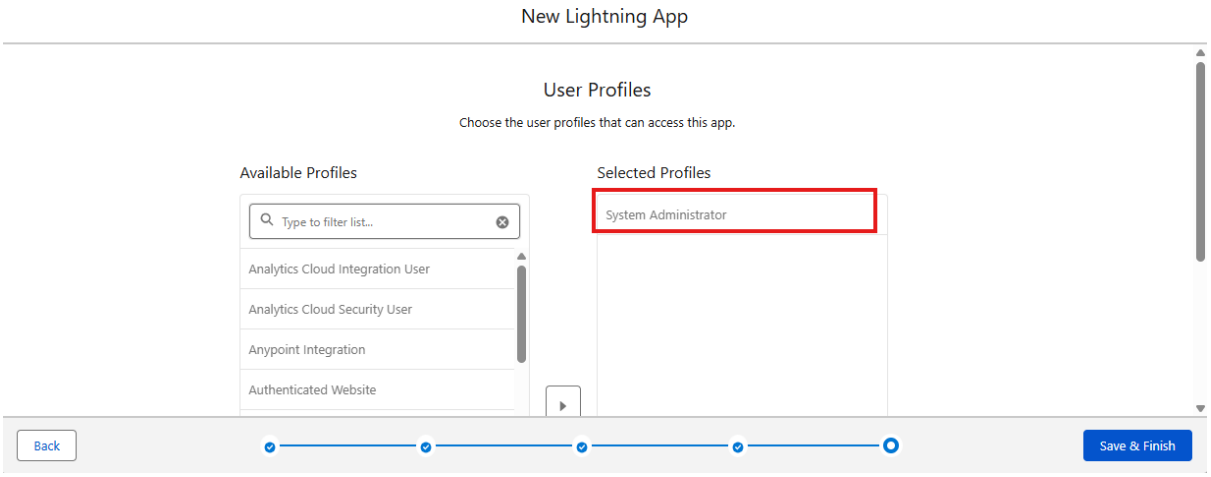
5. Navigation items

Select the created Custom Objects and required standard objects

- Players
- Match Performances
- Reports
- Dashboards



6. To Add User Profiles: System Administrator

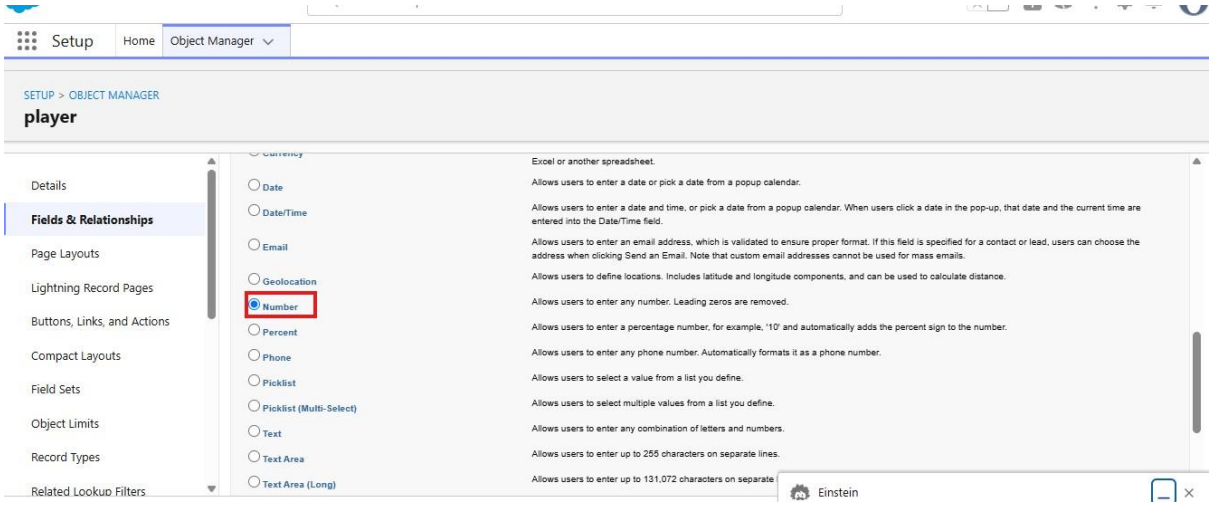


7.Click Save & Finish.

Milestone 5: Fields & Relationships

Activity 1: Create Custom fields for Player object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don’t see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created: **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn’t already exist.
6. Next, choose a data type as Text. Choosing a data type helps you format the field input.



7. Click on next.
8. Data Type: **Text**

   Field Label: Player Name

   Field Name: Auto Populates

   Length: 50

   Click on Next → Next → Save and new.
9. Repeat the Same steps for remaining fields

   - Data Type: **Text**

   Field Label: Player Number

   Field Name: Auto Populates

   Length: 3

   Click on Next → Next → Save and new.

   - Data Type: **Rich Text Area**

   Field Label: Profile Picture

   Field Name: Auto Populates

   Click on Next → Next → Save and new.

   - Data Type: **Date**

   Field Label: Date of Birth

   Field Name: Auto Populates .Click on Next → Next → Save and new.

   - Data Type: **Picklist**

   Field Label: Role

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

Batsman

Bowler

Batsman & Wicket Keeper

All Rounder

- Data Type: **Picklist**
  Field Label: Batting Style
  Field Name: Auto Populates
  Value: Enter value, with each value separated by a new line.
  Right- Handed
  Left - Handed
  Click on Next → Next → Save and new.

- Data Type: **Picklist**

Field Label: Bowling Style

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

Right-Arm Fast

Left-Arm Fast

Right-Arm Medium

Left-Arm Medium

Right-Arm Spin

Left-Arm Spin

Click on Next → Next → Save and new

- Data Type: **Picklist**

Field Label: Nationality

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line.

Australia

Bangladesh

England

India

New Zealand

Pakistan

South Africa

Sri Lanka

West Indies

Afghanistan

Zimbabwe

Ireland

Scotland

Netherlands

United Arab Emirates

Nepal

United States of America

Click on Next → Next → Save and new.

- Data Type: **Picklist**

Field Label: Team

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line

Chennai Super Kings (CSK)

Delhi Capitals (DC)

Kolkata Knight Riders (KKR)

Mumbai Indians (MI)

Punjab Kings (PBKS)

Rajasthan Royals (RR)

Royal Challengers Bangalore (RCB)

Sunrisers Hyderabad (SRH)

Gujarat Titans (GT)

Lucknow Supergiants (LSG)

- Data Type: **Formula**

Field Label: AVG Score

Field Name: Auto Populates

Formula Return: **Number**

Formula: IF (

Matches_Played__c > 0,

Total_Runs__c / Matches_Played__c ,

0

)



- Data Type: **Lookup Relationship**

Related To: Match Performance

Field Label: Match Performance

Field Name: Auto Populates

Click on Next → Next → Save.

Activity 2: Create Custom fields for Match Performance object.

1. Data Type: **Text**

Field Label: Match Performance Name

Field Name: Auto Populates

Length: 80

Click on Next → Next → Save and new.

2. Data Type: **Checkbox**

Field Label: Player of the Match

Field Name: Auto Populates

Click on Next → Next → Save and new.

3. Data Type: **Picklist**

Field Label: Match Result

Field Name: Auto Populates

Value: Enter value, with each value separated by a new line

WON

LOST

DRAW

Click on Next → Next → Save and new.

4. Data Type: **Number**

Field Label: Runs Scored

Field Name: Auto Populates

Click on Next → Next → Save and new.

5. Data Type: **Number**

Field Label: Balls Faced

Field Name: Auto Populates

Click on Next → Next → Save and new.

6. Data Type: **Number**

Field Label: Fours

Field Name: Auto Populates

Click on Next → Next → Save and new.

7. Data Type: **Number**

Field Label: Sixes

Field Name: Auto Populates

Click on Next → Next → Save and new.

8. Data Type: **Number**

Field Label: Overs Bowled

Field Name: Auto Populates

Click on Next → Next → Save and new.

9. Data Type: **Number**

Field Label: Wickets Taken

Field Name: Auto Populates

Click on Next → Next → Save and new.

10. Data Type: **Number**

Field Label: Runs Conceded

Field Name: Auto Populates

Click on Next → Next → Save and new.

11. Data Type: **Number**

Field Label: Maiden Overs

Field Name: Auto Populates

Click on Next → Next → Save and new.

12. Data Type: **Number**

Field Label: Stumpings

Field Name: Auto Populates

Click on Next → Next → Save and new

13. Data Type: **Number**

Field Label: Catches

Field Name: Auto Populates

Click on Next → Next → Save and new.

14. Data Type: **Master - Detail Relationship**

Related To: Player

Field Label: Player

Field Name: Auto Populates

Click on Next → Next → Save.

Milestone 6: Roll-up Summary

Activity 1: Create Roll-up Summary for Player object.

Note:- Roll-up Summary can only be created on parent object in Master-Detail Relationship.

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created: **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist. Click Next,
6. Choose a data type as **Roll-up Summary**. Click Next.

Field Label: Matches Played

Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: COUNT



Filter Criteria: All records should be included in the calculation.

Click on Next → Next → Save and new.

- Data Type: Roll-up Summary

Field Label: Number of 50's

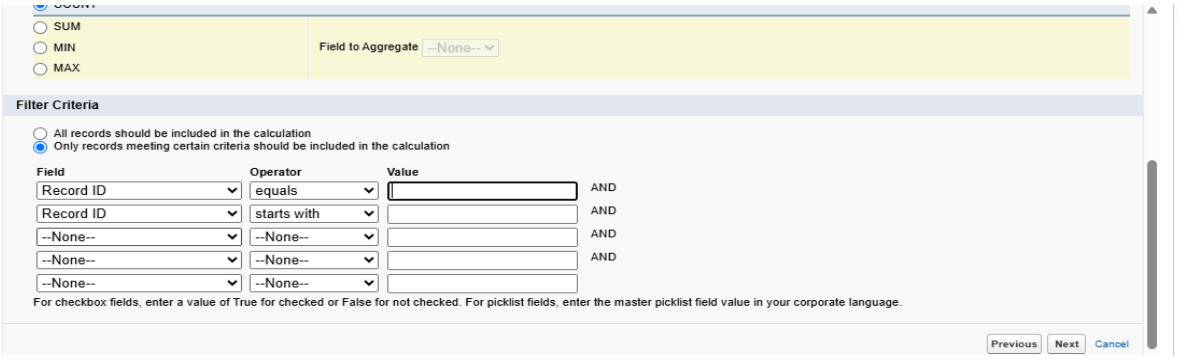
Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: COUNT

Filter Criteria: Only records meeting certain criteria should be included in the calculation.

Click on Next → Next → Save and new.



- Data Type: Roll-up Summary

Field Label: Number of 100's

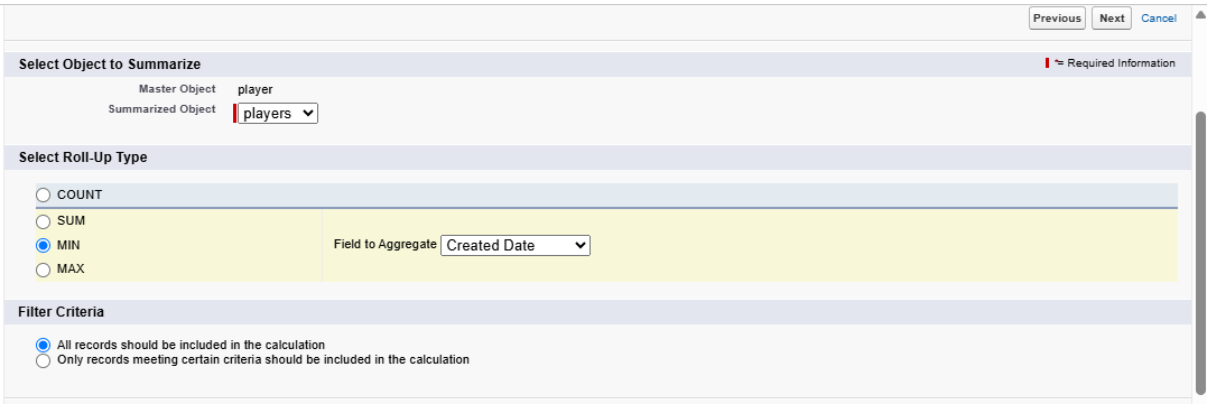
Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: COUNT

Filter Criteria: Only records meeting certain criteria should be included in the calculation.

Click on Next → Next → Save and new.



- Data Type: Roll-up Summary

Field Label: Total 4's

Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: SUM

Field To Aggregate: Fours

Filter Criteria: All records should be included in the calculation.

Click on Next → Next → Save and new.

- Data Type: Roll-up Summary

Field Label: Total 6's

Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: SUM

Field To Aggregate: Sixes

Filter Criteria: All records should be included in the calculation.

Click on Next → Next → Save and new.

- Data Type: Roll-up Summary

Field Label: Total Runs

Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: SUM

Field To Aggregate: Runs Scored

Filter Criteria: All records should be included in the calculation.

Click on Next → Next → Save and new.

- Data Type: Roll-up Summary

Field Label: Total Overs

Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: SUM

Field To Aggregate: Overs Bowled

Filter Criteria: All records should be included in the calculation.

Click on Next → Next → Save and new.

- Data Type: Roll-up Summary

Field Label: Total Wickets

Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: SUM

Field To Aggregate: Wickets Taken

Filter Criteria: All records should be included in the calculation.

Click on Next → Next → Save and new.

- Data Type : Roll-up Summary

Field Label: Total Stumpings

Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: SUM

Field To Aggregate: Stumpings

Filter Criteria: All records should be included in the calculation.

Click on Next → Next → Save and new.

- Data Type: Roll-up Summary

Field Label: Total Catches

Field Name: Auto Populates

Summarized Object: Match Performances

Select Roll-up Type: SUM

Field To Aggregate: Catches

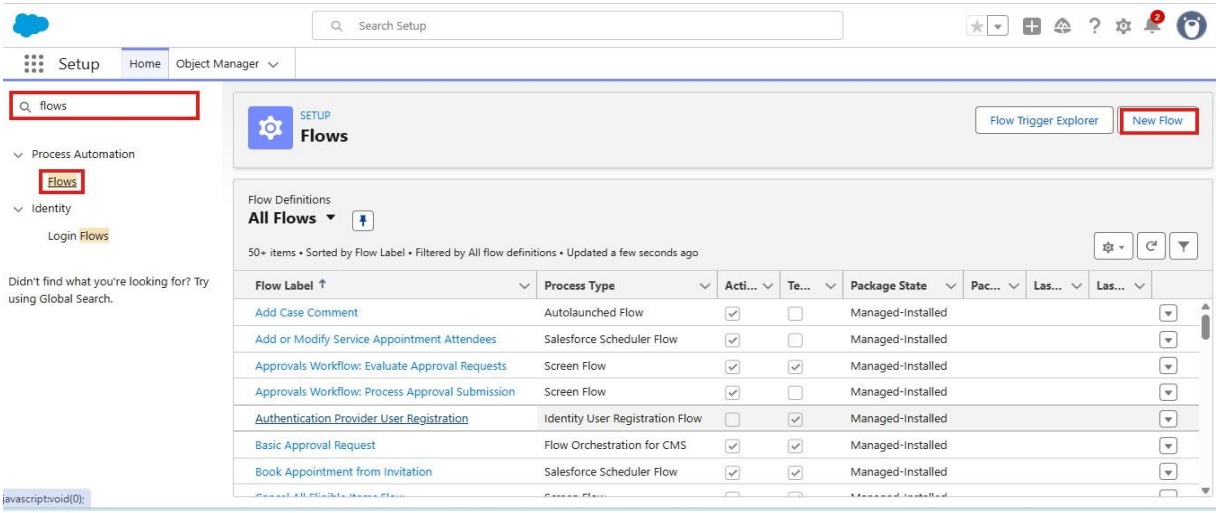
Filter Criteria: All records should be included in the calculation.

Click on Next → Next → Save and new.

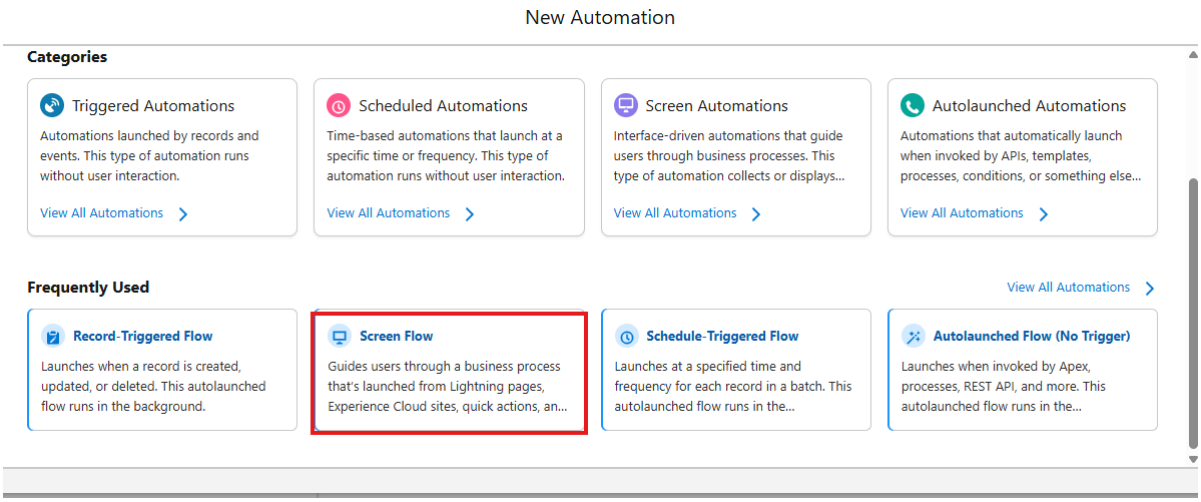
Milestone 7: Flows

Activity 1: Screen Flow

1. Go to setup, Type Flow in quick find box → Click on the Flow and Select the New Flow.



2. Select Screen Flow and click Create.



3. Click (+) Icon on the flow builder and Select **Screen Component.**

Label: Players screen

API: Players_screen

Select the **Picklist** component and place it on the screen.

Label: Team

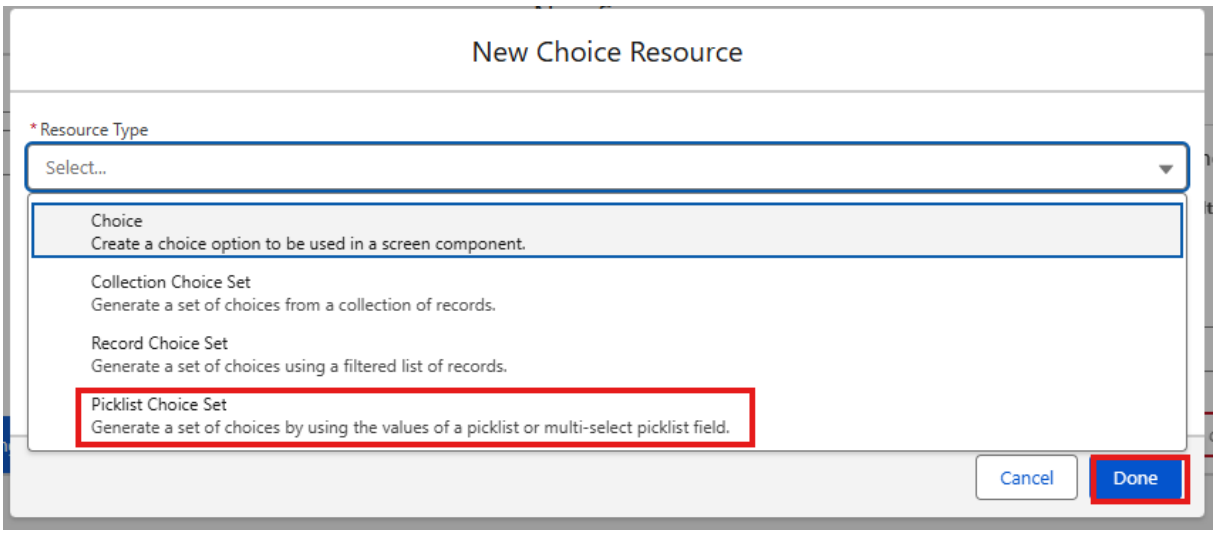
API: Team

From Component Pane on the left side select **Component Type**: Picklist

Data Type: Text

Choice: Click New Choice Resources

Resource Type: Picklist Choice Set



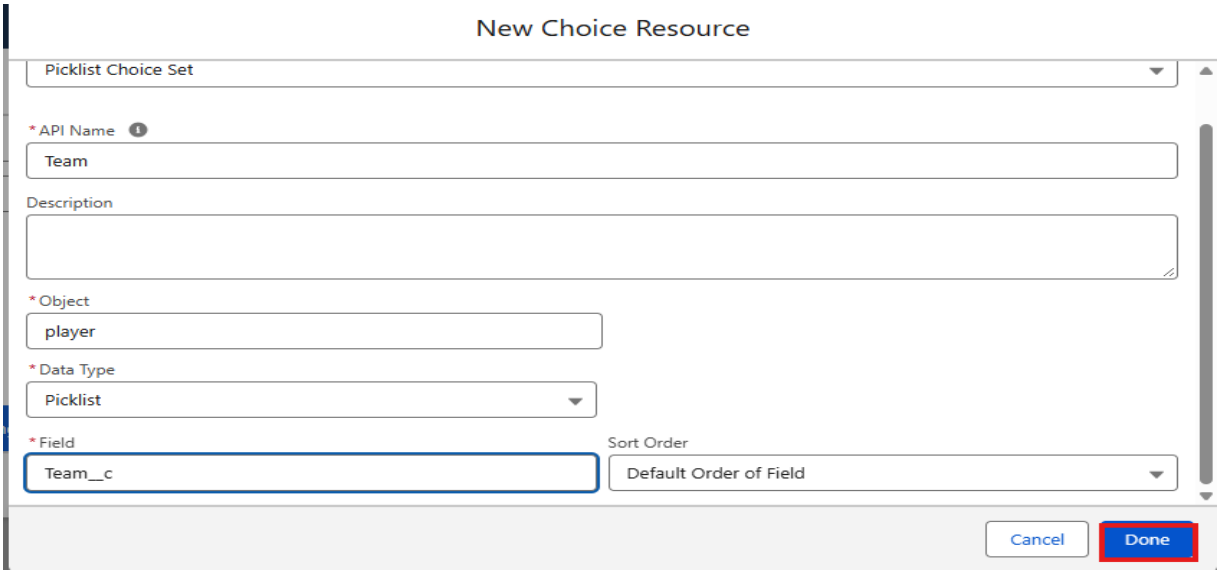
API Name: Team

Object: Player

Data Type: Picklist

Field: Team

Click Done.



4. Click (+) Icon on the flow builder and Select **Get Records Component.**

Label: Players

API Name: Players

Object: Player

Filter player records:

Field: Team

Operator: Equals

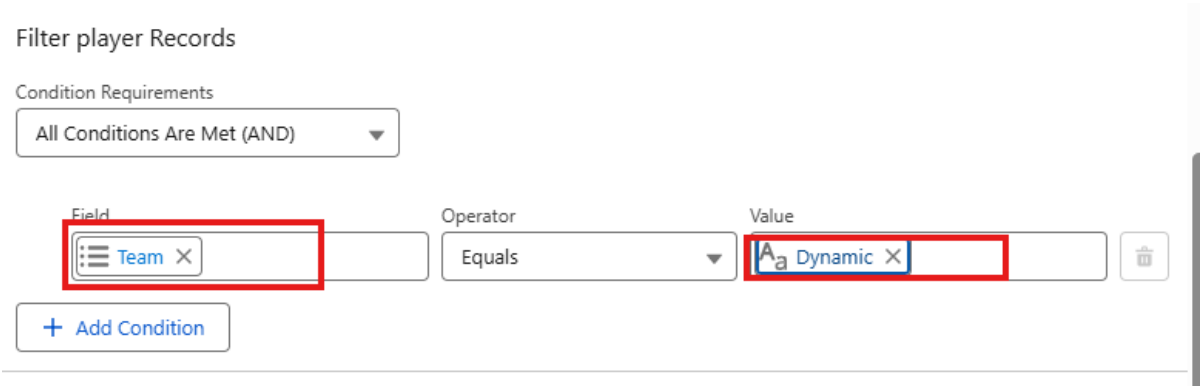
Value: Click New Resource

Resource Type: Variable

API Name: Dynamic

Data Type: Text

Default Value: Screen component: Team (Picklist)



5. Click (+) Icon on the flow builder and Select **Screen Component.**

Label: Player_info

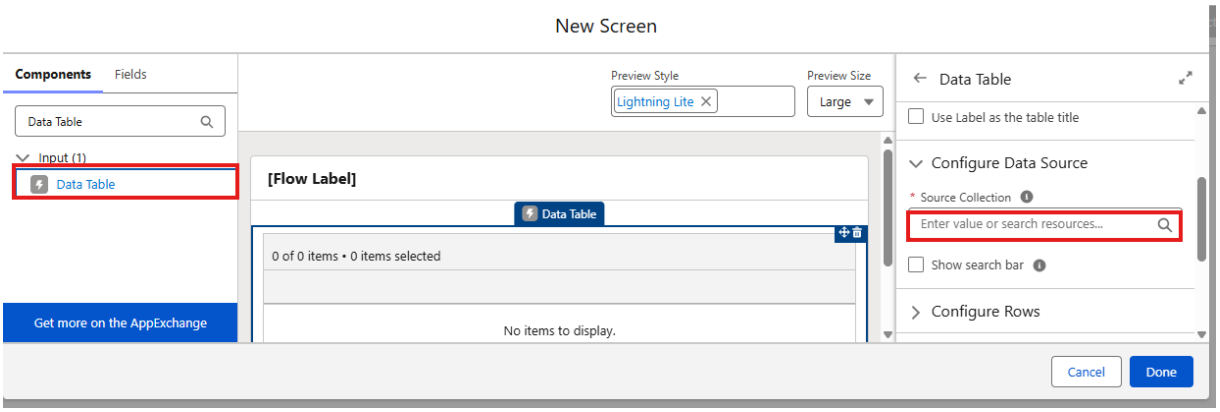
API Name: Player_info

From Component Pane on the left side select **Component Type:** Data table

API Name: Player_info_table

Label: Data Table

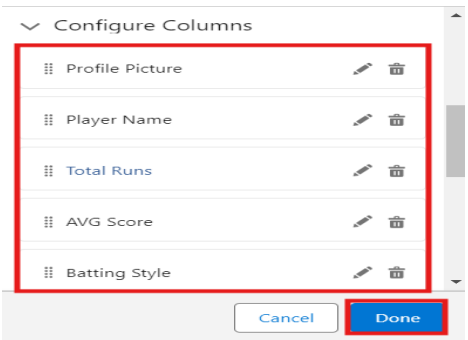
Configure Data Source: Source Collection: Players from players (Player__c)



In Configure Columns: Add all the required Columns

Profile Picture | Player Name | Total Runs | AVG Score | Batting Style | Bowling Style | Matches Played | Number of 50's | Number of 100's

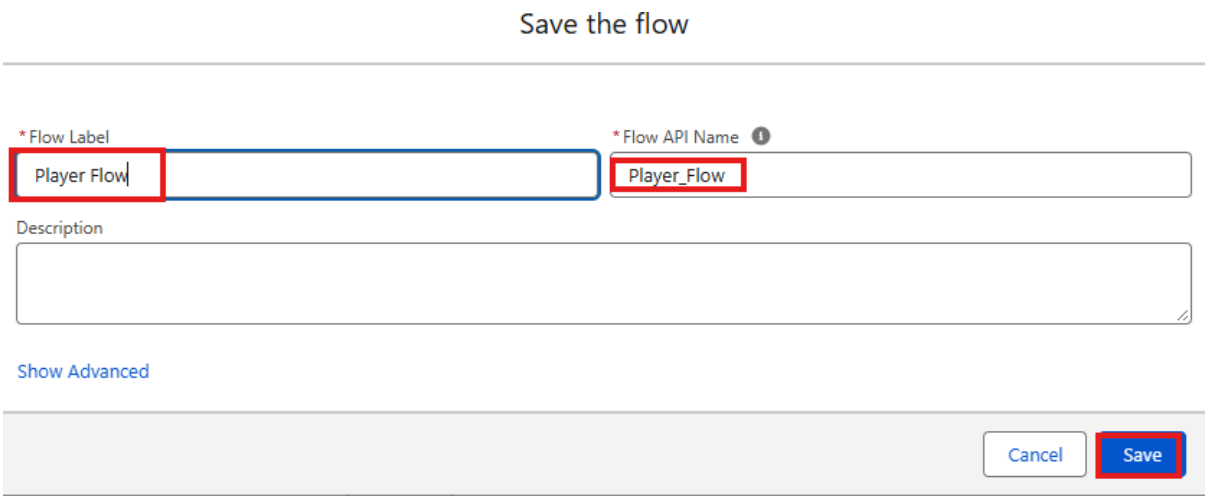
Click Done.

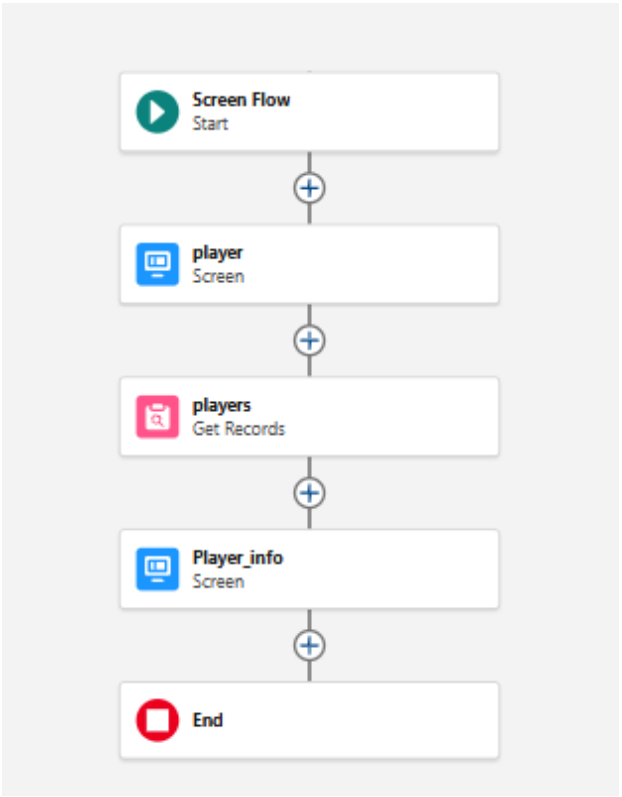


Click Save.

Flow Label: Player Flow

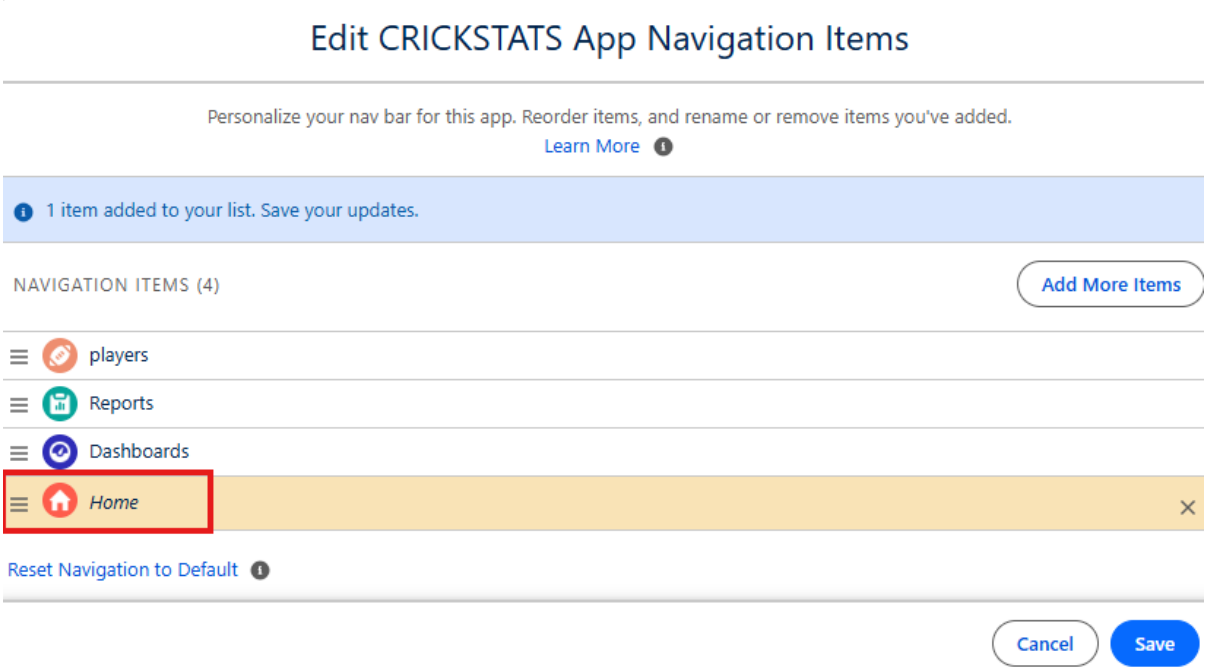
Flow API Name: Player_Flow



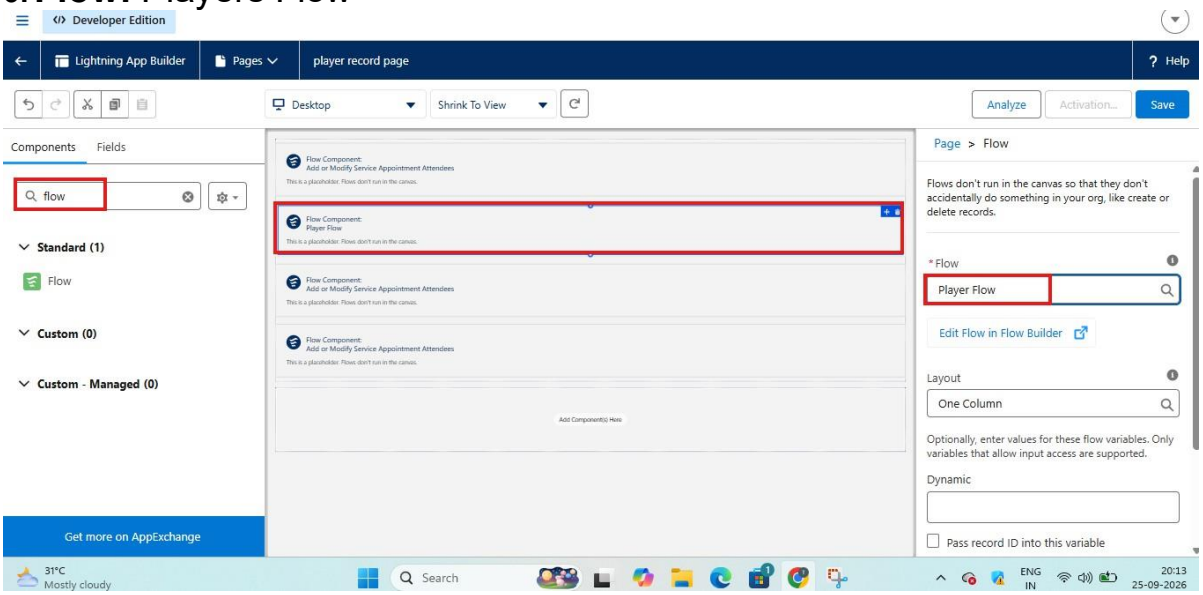


Activity 2: Flow on Home Page

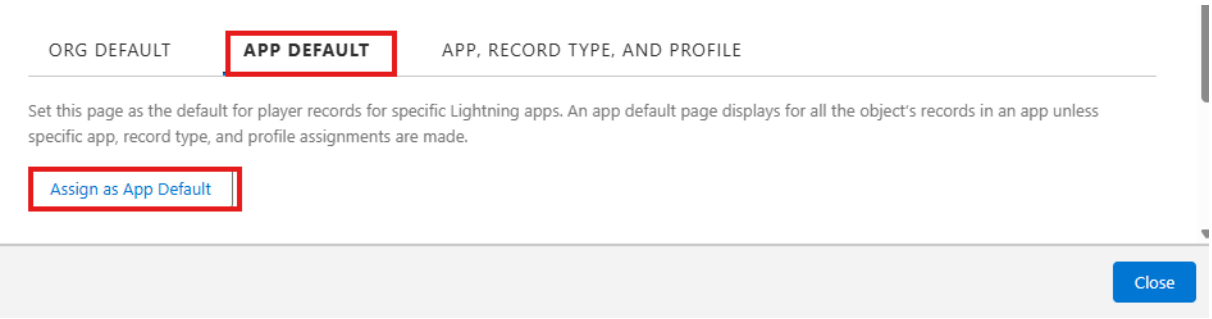
1. On your CRICKSTATS Application on the top right of the Navigation Bar click on **Personalize your nav bar icon (Pencil Icon)** and Add Home Page and click Save.



2. Remove all other components and Add **FLOW** component from the standard components on the left side.
3. **Flow:** Players Flow



4. Click **Save** and **Activation**.
5. Click on **App Default** and select **Assign to Apps.**



6. Select **CRICKSTAS** App and Click **Next** and then click **Save**.

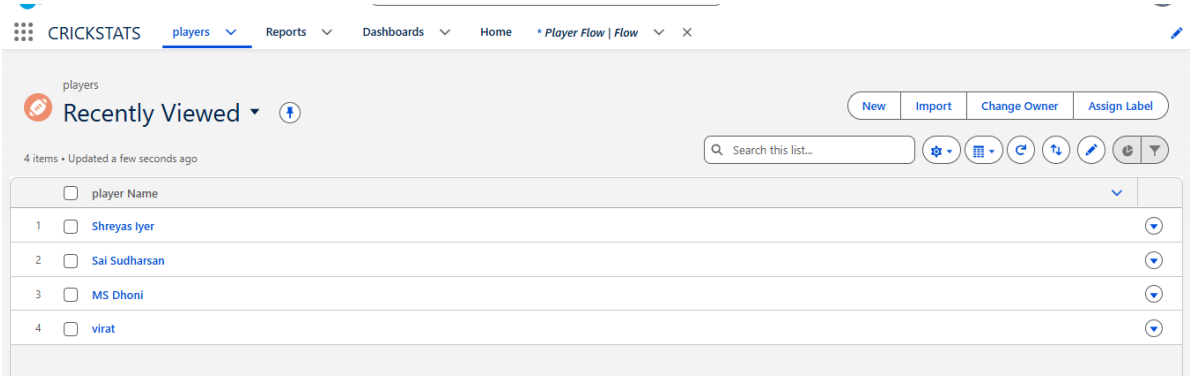
7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce Lightning user interface to verify the correct functioning of all implemented features in the CRICKSTATS system. These include record creation for Player and Match Performance objects; validation rule checks for mandatory fields, score limits, and input constraints; verification of record type behavior for Domestic and International match categories; and flow execution for automatic updating of player statistics and data consistency. The functionality of roll-up summary fields was also tested to ensure accurate calculation of total runs, wickets, averages, and performance metrics.

In addition, system performance was evaluated by monitoring response time during data entry, report generation, and dashboard loading. The application was tested under multiple user interactions to ensure stable performance without delays. Screenshots of generated reports and interactive dashboards were documented to confirm accurate data visualization, real-time updates, and reliable performance tracking. These testing activities ensured that the CRICKSTATS platform met both functional and performance requirements and was ready for deployment.

NOTE: Enter some records in the Player Object and Match Performance Object to get the results.

1. Select one Team from the Picklist and Click **Next**. Then give the information. Next to save it.



7.ADVANTAGES & DISADVANTAGES

Advantages

- Centralized system
- Real-time analytics
- Secure platform
- Scalable design
- Reduced manual work

Disadvantages

- Requires Salesforce license
- Internet dependency
- Limited offline access

8.Conclusion:

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making.

The project demonstrates practical application of SDLC, Agile methodology, and CRM-based development.

9. FUTURE SCOPE

- Mobile application integration
- AI-based performance prediction
- Video analysis integration
- Player ranking system
- Public statistics portal
- Multi-league support

10. APPENDIX

- Salesforce Objects & Fields
- Manual match statistics
- Flow Configurations

Thank You